

In the case of witnessed tests, all test records shall be initialled by representatives of the parties witnessing the test, each of whom shall be provided with a copy of all records.

The test results shall be evaluated to the extent possible, while the tests are in progress. In order that questionable measurements can be re-evaluated, it is advisable that the installation and instrumentation remain intact until accurate data are obtained.

If required, the test results shall be summarized in a report. Further guidance regarding the contents of a test report and a suitable pump test sheet is given in Annex F.

5.6 Test arrangements

The conditions necessary to ensure satisfactory measurement of the characteristics of operation are defined here, taking into account the measurement uncertainty required for tests of grade 1, 2 and 3.

The performance of a pump in a given test arrangement, however accurately measured, cannot be assumed to be a correspondingly accurate indication of its performance in another arrangement. Recommendations and general guidance about suitable pipe arrangements to ensure satisfactory measurements for flow and head are given in Annex A and, if necessary, they can be used in conjunction with the International Standards on measurement of flow rates in closed conduits concerning the different methods (see D.3).

5.7 Test conditions

5.7.1 Test procedure

The duration of the test shall be sufficient to obtain repeatable results.

All measurements shall be carried out under steady-state conditions (see 4.3.2 and Table 3). If not otherwise specified, the tests shall be conducted under conditions where cavitation does not affect the performance of the pump.

A minimum of five test points shall be taken for all performance tests, regardless of acceptance level, with one of the points being within -5% and 0% and one being within 0% and $+5\%$ of the guarantee point flow rate. The other three points shall be spaced out over the allowable operating range of the pump performance curve, with points taken near the maximum allowable head and flow regions.

NOTE Other test procedures apply to NPSH tests; see 5.8.

5.7.2 Speed of rotation during test

Unless otherwise agreed, tests may be carried out at a speed of rotation within the range 50% to 120% of the specified speed of rotation to establish rate of flow, pump total head and power input. In the case the variation of speed is within 20% of the specified speed, the efficiency change is considered negligible.

For NPSH tests, the speed of rotation should lie within the range 80% to 120% of the specified speed of rotation, provided the rate of flow lies within 50% and 120% of the rate of flow corresponding to the maximum efficiency at the test speed of rotation.

5.8 NPSH tests

5.8.1 General

5.8.1.1 Objective of NPSH tests

The objective of the NPSH test is to verify the pump's NPSHR for the agreed guarantee. This test deals only with measurements relating to the hydraulic performance of the pump (variations of head, flow, power) and not with other effects, which can be caused by cavitation (e.g. noise, vibration, erosion).

Cavitation effects can be detected as a drop in head or power at a given rate of flow. In the case of multistage pumps, the head drop shall be relative to the head of the first stage, which should be measured if accessible. For very low head pumps, a head drop larger than 3 % may be agreed upon.

In most cases, cavitation tests are conducted with clean, cold water. Cavitation tests in water cannot accurately predict the behaviour of the pump with liquids other than clean, cold water.

Air content can have a significant effect on measured NPSHR-values and shall be considered.

5.8.2 NPSH test types

5.8.2.1 Type I test — Determination of NPSH3 for multiple flow rates

In this test, NPSH is reduced progressively until the drop of the total head at constant flow rate reaches 3 %. This value of NPSH is NPSH3 (see Table 10). A minimum of four different suitably spaced flow rates shall be evaluated within the allowable operation range.

5.8.2.2 Type II test — Determination of NPSH3 for a single flow rate

In this test, NPSH is reduced progressively until the drop of the total head of 3 % at constant flow rate can be determined. This value of NPSH is NPSH3 (see Table 10).

5.8.2.3 Type III test — Verification of limited influence of cavitation on the performance at specified NPSHA

Verification is carried out at the specified NPSHA to show that the hydraulic performance of the pump is not affected by cavitation more than 3 % drop of total head.

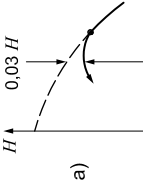
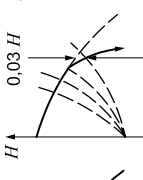
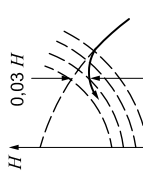
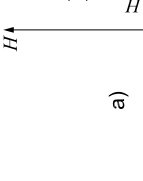
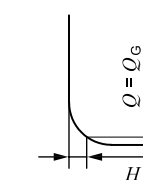
5.8.2.4 Type IV — Verification of guaranteed characteristics at specified NPSHA

The pump meets the requirements if the guaranteed pump total head and power are obtained according to 4.4, under the specified rate of flow and under the specified NPSHA.

5.8.2.5 Tolerance factor for NPSHR

The measured NPSHR value shall not exceed the guaranteed NPSHR value.

Table 10 — Methods of determining NPSH3

Type of installation	Open sump	Open sump	Open sump	Open sump	Closed loop	Closed loop	Closed loop	Closed sump or loop
Independent variable	Inlet throttle valve	Outlet throttle valve	Water level	Inlet throttle valve	Water level	Pressure in the tank	Temperature (vapour pressure)	Temperature (vapour pressure)
Constant	Outlet throttle valve	Inlet throttle valve	Inlet and outlet throttle valves	Rate of flow	Rate of flow	Rate of flow	Rate of flow	Inlet and outlet throttle valves
Quantities that vary with changes to the independent variable	Total head, rate of flow, NPSHA, water level	Total head, rate of flow, NPSHA, water level	Total head rate of flow NPSHA	NPSHA, total head; outlet throttle valve (for constant rate of flow)	NPSHA, total head; outlet throttle valve	Total head, NPSHA, outlet throttle valve (for constant rate of flow; when total head begins to drop)	NPSHA, head, outlet throttle valve (for constant rate of flow, when head begins to drop)	— — — total head and rate of flow, when a certain level of cavitation is reached
Head characteristic curve versus rate of flow and NPSH								
NPSH characteristic curve versus rate of flow	